

Model Benchmark Report

A comprehensive record of AI model testing for automated square footage extraction from commercial real estate documents.

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API Platform	OpenRouter
Total API Calls	1,015
Total Spend	\$1.76
Models Tested	25

Project Summary

This report documents the complete benchmarking process for an automated lease document OCR pipeline built for ChainXY. The pipeline ingests commercial real estate PDFs — including formal leases, marketing flyers, site plans, and tenant rosters — and extracts the primary square footage figure using AI language models via the OpenRouter API.

The testing covered 25 different AI models across multiple prompt styles, effort levels, extraction modes, temperatures, and reasoning configurations, totalling 1,015 API calls and \$1.76 in spend.

25	1,015	\$1.76	203
Models Tested	Total API Calls	Total Spend	PDFs Processed

Pipeline Architecture

The pipeline processes each PDF through the following stages:

- 1. PDF Ingest** — PyMuPDF (fitz) opens the file, validates page count (minimum 1 page), and prepares it for processing.
- 2. Page Classification** — Each page is classified as searchable (extractable text) or scanned (image-only) using a 100-character non-whitespace threshold.
- 3. Text / OCR Extraction** — Searchable pages are read directly. Scanned pages are rasterised at 200 DPI and processed through Tesseract OCR. Pages are filtered by keywords (GLA, sq ft, demised premises, etc.) to find candidates.
- 4. AI Extraction** — Candidate text is sent to an LLM via OpenRouter with a structured prompt requesting square footage, unit, confidence, and evidence snippet as raw JSON.
- 5. Output** — Results are appended to an enrichment CSV (store_id, filename, square_footage, unit, confidence, evidence_snippet).

Production Run Results

A full batch was run on 203 PDFs from the SiteFiles dataset using Claude Sonnet 4.5 with the relaxed prompt:

Total PDFs	Successful	Skipped	Failed
203	176	27	0
100%	87%	13% (no candidate text)	0%

Variables Under Test

Six independent variables were tested in combination. Every run logs the model, prompt, effort level, extraction mode, temperature, reasoning flag, extracted value, confidence, cost, and latency into a timestamped CSV.

Variable	Options Tested	Finding
Model	25 models across 10+ families	Biggest impact on accuracy and cost
Prompt Style	relaxed · strict · few-shot	Strict = near-zero on marketing docs; relaxed = best
Effort Level	first2 · half · full	first2 matches full accuracy on this dataset
Extraction Mode	text · vision	Text sufficient; vision untested at full scale
Temperature	0.0 · 0.3	No meaningful difference found
Reasoning	off · on	Same or worse accuracy, 2-3x slower and costlier

Prompt Styles

Relaxed (production) [relaxed]

Accept any square footage figure: total center size, GLA, demised premises, suite size, or building size. Marketing flyers and tenant rosters are valid sources. Return highest/most prominent figure.

Strict (lease-only) [strict]

Only extract from formal lease clauses: Demised Premises, Premises, GLA, Rentable Area. Do NOT use marketing text, site plans, or tenant rosters. Result: near-zero extraction rate on this dataset (documents are mostly flyers).

Few-shot (examples) [few_shot]

Includes three worked examples before the task. Slight accuracy benefit over relaxed on some models, but adds token cost. No net advantage found in this dataset.

Full Model Comparison

All models tested with the relaxed prompt, temperature 0, no reasoning, effort level: first2 pages. Sorted by accuracy then cost. Models marked with * have fewer than 15 runs — data is preliminary.

Model	Family	Extracted	High Conf	Cost / 15 PDFs	Avg Speed	Runs
Qwen 2.5 72B *	Alibaba	2 / 2	2 / 2	\$0.00784	1.7s	2
Mistral Nemo	Mistral	15 / 16	15 / 16	\$0.00105	1.7s	16
Claude Sonnet 4.5	Anthropic	19 / 21	19 / 21	\$0.06773	2.3s	21
Gemma 3 12B	Google	13 / 15	12 / 15	\$0.00045	1.5s	15
Gemma 3n E4B	Google	13 / 15	12 / 15	\$0.00047	2.1s	15
DeepSeek Chat	DeepSeek	13 / 15	13 / 15	\$0.00215	2.6s	15
GPT-4o Mini	OpenAI	13 / 15	13 / 15	\$0.00242	1.1s	15
GPT-4o	OpenAI	13 / 15	12 / 15	\$0.04036	0.9s	15
Gemini 2.5 Flash Lite	Google	13 / 16	13 / 16	\$0.00099	0.6s	16
MiniMax	MiniMax	13 / 16	13 / 16	\$0.00283	1.9s	16
Gemini 2.5 Flash	Google	13 / 16	13 / 16	\$0.00402	0.8s	16
Qwen 3 235B	Alibaba	13 / 16	12 / 16	\$0.00686	6.1s	16
Phi-4	Microsoft	12 / 15	12 / 15	\$0.00049	0.9s	15
Llama 4 Scout	Meta	12 / 15	12 / 15	\$0.00072	0.6s	15
GPT-4.1 Nano	OpenAI	12 / 15	12 / 15	\$0.00082	1.0s	15
Gemma 3 27B	Google	12 / 15	12 / 15	\$0.00087	2.3s	15
Llama 4 Maverick	Meta	12 / 15	12 / 15	\$0.00154	1.3s	15
GPT-4.1 Mini	OpenAI	12 / 15	11 / 15	\$0.00325	1.4s	15
Qwen 2.5 VL 72B	Alibaba	12 / 16	12 / 16	\$0.00314	1.9s	16
Kimi K2	Moonshot	12 / 16	11 / 16	\$0.00731	4.0s	16
Cohere Command R7B	Cohere	11 / 15	10 / 15	\$0.00037	0.9s	15
Llama 3.3 70B	Meta	11 / 15	11 / 15	\$0.00182	2.6s	15
Mistral Large	Mistral	11 / 15	10 / 15	\$0.01756	1.3s	15

Model	Family	Extracted	High Conf	Cost / 15 PDFs	Avg Speed	Runs
Qwen 3 30B	Alibaba	10 / 16	10 / 16	\$0.00243	3.3s	16
Gemini 2.5 Pro	Google	0 / 16	0 / 16	\$0.07891	5.2s	16

* Fewer than 15 PDFs tested — preliminary result. Cost / 15 PDFs is extrapolated for models with more or fewer runs.

Prompt Style Impact

Tested on Claude Sonnet 4.5, 15 PDFs, temperature 0, no reasoning:

Prompt Style	Extracted	High Conf	Notes
relaxed	13 / 15	13 / 15	Best — accepts any SF figure
few_shot	13 / 15	13 / 15	Same as relaxed but higher token cost
strict	1 / 15	1 / 15	Only formal lease clauses — useless on flyers

Reasoning Mode Impact

Claude extended thinking vs standard, relaxed prompt, temperature 0, 15 PDFs:

Mode	Extracted	Avg Cost / PDF	Avg Latency	Verdict
off (standard)	13 / 15	\$0.0044	2.0s	Use this
on (extended thinking)	13 / 15	\$0.0079	5.3s	Same accuracy, 1.8x cost, 2.7x slower

DeepSeek Reasoning (R1) Impact

Mode	Extracted	Avg Latency	Verdict
off (deepseek-chat)	13 / 15	2.6s	Best — accurate and fast
on (deepseek-r1 swap)	8 / 15	30-100s	Significantly worse, very slow

Summary of Results

Mistral Nemo is the accuracy leader

14-15/15 PDFs extracted correctly at \$0.00007/call average — a tiny, cheap model that outperforms Claude Sonnet 4.5 on this specific task. Recommended for production if this accuracy holds on a larger sample.

Gemma 3 12B is the best cost-accuracy tradeoff

13/15 accuracy at \$0.00003/call — 135x cheaper than Claude Sonnet 4.5 (\$0.0045/call) for the same extraction rate. Strong candidate for high-volume production use.

Gemini 2.5 Flash Lite is the fastest and nearly free

12-13/15 accuracy, 0.6s average latency, \$0.00007/call. Best option if throughput and speed are the priority.

Relaxed prompt is mandatory for this dataset

The strict prompt (formal lease clauses only) produces near-zero extraction on marketing flyers and site plans — which make up the majority of the dataset. Always use the relaxed prompt.

Reasoning modes add cost with no accuracy benefit

Extended thinking (Claude) and DeepSeek R1 produce identical or worse accuracy at 2-3x the cost and latency. Disable reasoning for this task.

Temperature has zero effect

T=0.0 and T=0.3 produced identical results across all models. Lock at 0.

Gemini 2.5 Pro is not suitable for this task

0/15 extraction rate across a full 15-PDF run (\$0.08 total). Consistently returns null with low confidence. The model may need a very different prompt format or more pages to perform.

Baidu ERNIE not available on OpenRouter

All Baidu model IDs return 400 errors on this platform. Would require a direct integration with the Baidu Qianfan API — a separate project.

First-2-pages effort matches full-document accuracy

The effort level (how much text to send) had no impact on accuracy for this dataset — square footage figures appear early in documents. Use first2 in production to minimise token cost.

Vision extraction implemented but not fully benchmarked

Claude Vision and other vision-capable models can process rasterised page images. Full vision benchmarking across 15+ PDFs is pending.

Next Steps

Priority	Action	Rationale
	Run Mistral Nemo on 50+ PDFs	Confirm 14/15 result holds at scale before recommending as production model
High	Run Gemma 3 12B on 50+ PDFs	Validate 13/15 result — already cheapest model at that accuracy tier
	Full vision benchmark	Test vision extraction on the scanned-heavy PDFs where Tesseract OCR struggles
Medium	LlamaParse integration	Dedicated PDF parser (cloud.llamaindex.ai) — separate API key required. Strong on
	Connect Tier 1 + Tier 3 outputs	Merge AI-extracted SF figures with Overture Maps building footprint data into a single
Low	Baidu ERNIE direct integration	Would require Baidu Qianfan API key — separate project outside OpenRouter

Cost Projection

Based on benchmark results, estimated cost to run extraction on 1,000 PDFs:

Model	Cost / 1,000 PDFs	Accuracy (est.)	Notes
Mistral Nemo	\$0.006	~93%	New leader — needs validation
Gemma 3 12B	\$0.030	~87%	Best validated cheap option
Gemini 2.5 Flash Lite	\$0.007	~80%	Fastest, nearly free
GPT-4o Mini	\$0.161	~87%	Previous benchmark — well established
Claude Sonnet 4.5	\$4.50	~87%	Original model — gold standard